GBFS Programs

1

```
import heapq

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [], 'E': [], 'F': [], 'G': []
}

h = {
    'A': 6, 'B': 4, 'C': 2,
    'D': 5, 'E': 3, 'F': 1, 'G': 0
}

def gbfs(start, goal):
    visited = set()
    pq = [(h[start], start)]

    while pq:
        _, node = heapq.heappop(pq)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)
```

```
    if node == goal:
        print("\nGoal reached")
        return

    for child in graph[node]:
        if child not in visited:
            heapq.heappush(pq, (h[child], child))
```

gbfs('A', 'G')



2

```
import heapq

graph = {
    'S': ['A', 'B'],
    'A': ['C', 'D'],
    'B': ['E', 'F'],
```

```
    'C': [], 'D': [], 'E': [], 'F': []
}

h = {
    'S': 6, 'A': 4, 'B': 2,
    'C': 5, 'D': 3, 'E': 0, 'F': 1
}

def gbfs(start, goal):
    visited = set()
    pq = [(h[start], start)]

    while pq:
        _, node = heapq.heappop(pq)

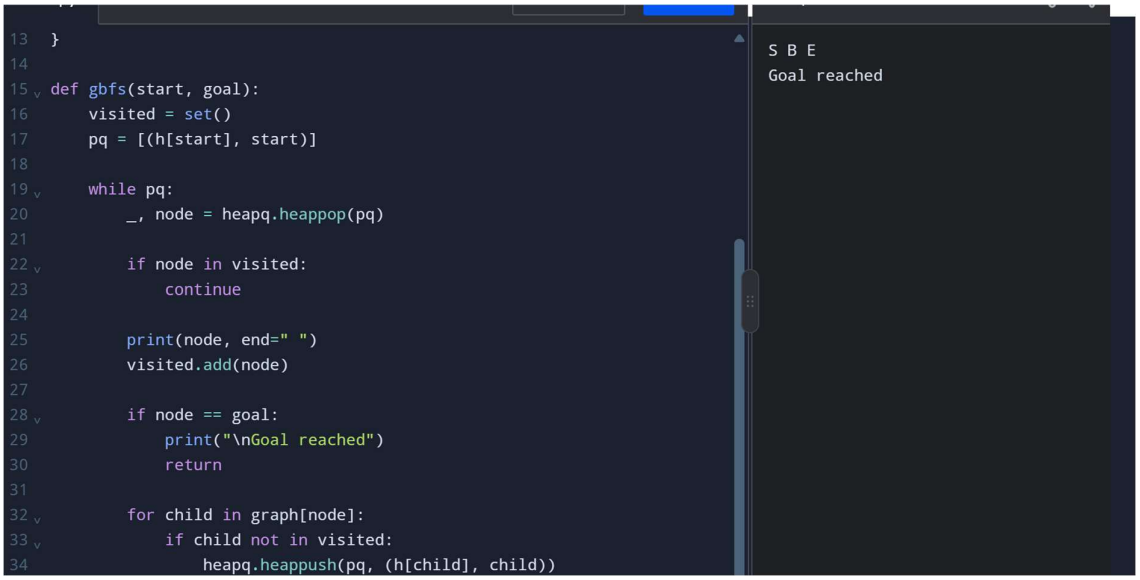
        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal reached")
            return

        for child in graph[node]:
            if child not in visited:
                heapq.heappush(pq, (h[child], child))
```

gbfs('S', 'E')



3

import heapq

graph = {

'A': ['B', 'C'],

'B': ['D', 'E'],

'C': ['F'],

'D': [], 'E': [], 'F': []

}

h = {

'A': 5, 'B': 3, 'C': 4,

'D': 0, 'E': 2, 'F': 1

}

def gbfs(start, goal):

```
    visited = set()
    pq = [(h[start], start)]

    while pq:
        _, node = heapq.heappop(pq)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal reached")
            return

        for child in graph[node]:
            if child not in visited:
                heapq.heappush(pq, (h[child], child))

gbfs('A', 'D')
```



```
def gbfs(start, goal):
    visited = set()
    pq = [(h[start], start)]

    while pq:
        _, node = heapq.heappop(pq)

        if node in visited:
            continue

        print(node, end=" ")
        visited.add(node)

        if node == goal:
            print("\nGoal reached")
            return

        for child in graph[node]:
            if child not in visited:
                heapq.heappush(pq, (h[child], child))
```

```
A B D
Goal reached
```

4

```
import heapq


graph = {
   'S': ['A', 'B'],
   'A': ['C', 'D'],
   'B': ['E', 'F'],
   'C': [], 'D': [], 'E': [], 'F': []
}


h = {
   'S': 7, 'A': 5, 'B': 3,
   'C': 6, 'D': 4, 'E': 2, 'F': 0
}


def gbfs(start, goal):
   visited = set()
   pq = [(h[start], start)]


   while pq:
```

```
    _, node = heapq.heappop(pq)

    if node in visited:
        continue

    print(node, end=" ")
    visited.add(node)

    if node == goal:
        print("\nGoal reached")
        return

    for child in graph[node]:
        if child not in visited:
            heapq.heappush(pq, (h[child], child))
```

gbfs('S', 'F')

